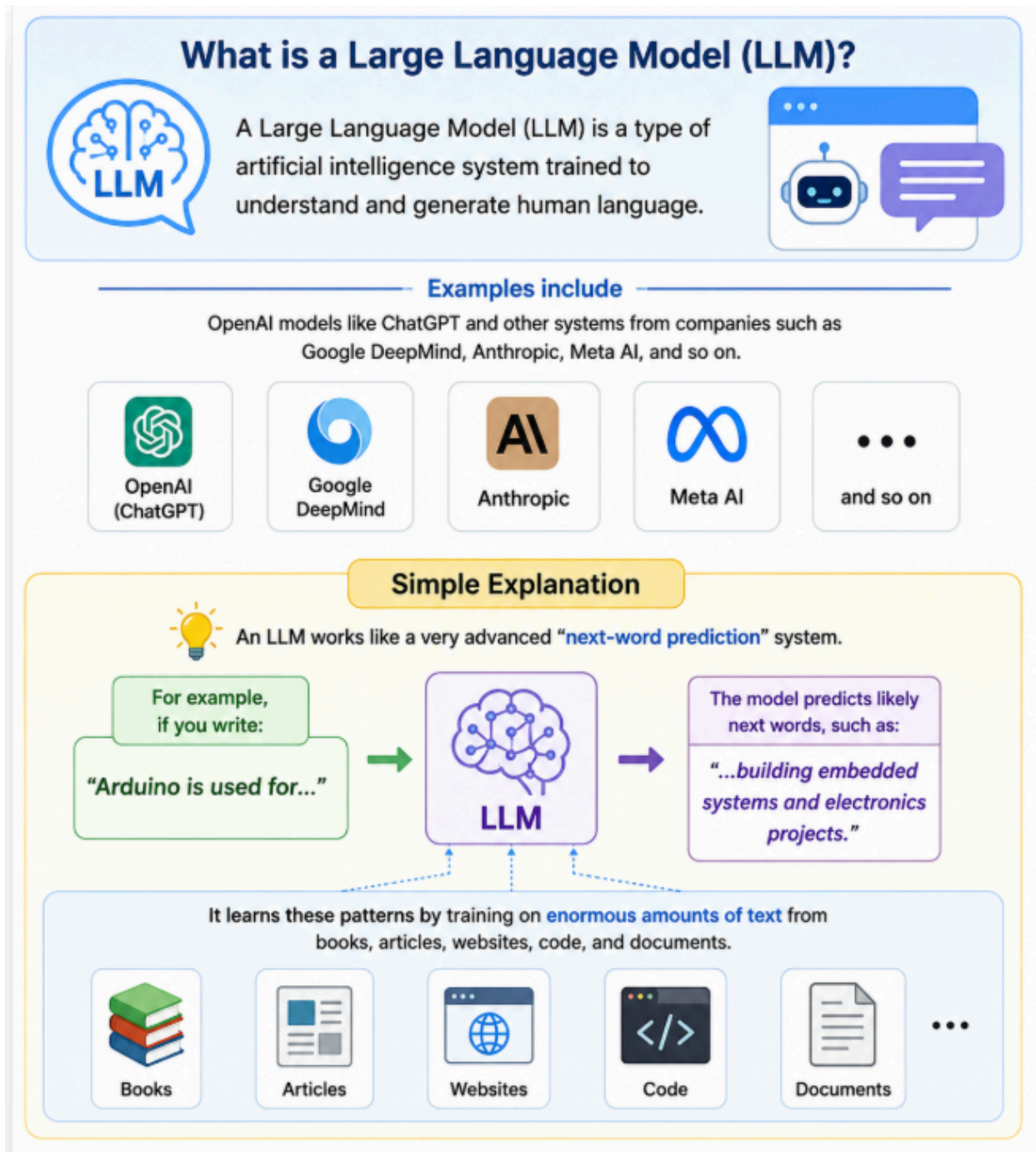


What is a Large Language Model (LLM)?



A **Large Language Model (LLM)** is a type of **artificial intelligence system** trained to understand and **generate human language**.

Examples include OpenAI models like ChatGPT and other systems from companies such as [Google DeepMind](#), [Anthropic](#), [Meta AI](#), and so on.

Simple Explanation

An LLM works like a very advanced “next-word prediction” system.

For example, if you write: “Arduino is used for...”

The model predicts likely next **words**, such as "...[building embedded systems and electronics projects](#)."

It learns these patterns by training on enormous amounts of text from books, articles, websites, code, and documents.

Why is it called “Large”?

“**Large**” refers to:

- Huge training datasets
- Billions or trillions of parameters (internal mathematical weights)
- Massive computing power is used during training.

Some modern LLMs are trained on thousands of GPUs and petabytes of text data.

How an LLM Works

At a high level:

1. Text is converted into numbers called **tokens**.
2. The model analyzes relationships between **words**.
3. It predicts the most probable **next token**.
4. Repeating this process generates sentences, paragraphs, or code.

Most modern **LLMs** use a **neural network** architecture called the [transformer](#).

The transformer was introduced in the famous 2017 paper "**Attention Is All You Need**."

Core Abilities of LLMs

LLMs can:

- Answer questions

- Write essays and reports.
- Generate code
- Translate languages
- Summarize documents
- Create lesson plans
- Help with cybersecurity analysis
- Assist with Arduino and Raspberry Pi programming
- Build chatbots and dashboards

Example

If you ask:

“Write Python code for packet sniffing”

an LLM can generate something like:

```
</> Python

from scapy.all import sniff

def packet_callback(packet):
    print(packet.summary())

sniff(prn=packet_callback, count=10)
```

Types of Language Models

Traditional Language Models

Older statistical systems:

- N-grams
- Markov models

Modern Large Language Models

Deep learning-based:

- GPT

- Llama
- Gemini
- Claude

Mathematical Idea Behind It

LLMs estimate probabilities like:

$$P(w_t | w_1, w_2, \dots, w_{t-1})$$

This means:

“What is the probability of the next word given all previous words?”

Important Limitations

LLMs:

- Can make mistakes (“hallucinations”).
- Do not truly “understand” as humans do.
- May produce biased or outdated information.
- Need verification for critical tasks.

Hugging Face: What is an LLM?

<https://huggingface.co/learn/agents-course/en/unit1/what-are-llms>